User Guide for OpenFOAM-8

Porous-media modeling of catalytic packed-bed reactors with detailed surface chemistry in OpenFOAM: modeling and validation

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1. General descriptions

The framework is fully native OpenFOAM(OF) and comprises newly developed libraries and solvers as follows:

- libraries:
  - **thermophysicalModels** includes surface chemistry sub-library.
- solvers:
  - **porousRhoPBRFoam**
  - **porousPBRFoam**
- utilities:
  - **surfChemkinToFoam**

1.1. Library for surface chemistry in OpenFOAM

Figure 1 presents the class diagram of the updated `thermophysicalModels` library, including the `surfaceChemistryModels` sub-library used for heterogeneous surface chemistry. The thermophysical library developed in this work extends the `surfaceChemistryFoam` framework [1] to support porous-media simulations of catalytic packed-bed reactors. The resulting class structure is presented in Fig. 1. Detailed descriptions of the classes inherited from `surfaceChemistryFoam` are provided in *surfaceChemistryFoam_ userGuide_ OF8.pdf*, available in the `porousPBRFoam-8/documentations` directory [2]. The present section therefore focuses on the classes introduced for porous-media modeling.

- *PorousMediaSurfaceChemistryModel* incorporates detailed heterogeneous surface chemistry into the porous-media formulation. It solves the coupled cell-local ODE systems for the gas-phase and surface species in each porous cell. The resulting surface-reaction rates are scaled by the surface-to-volume ratio to obtain the volumetric species-production and heat-release source terms used in the macroscopic species and energy conservation equations.

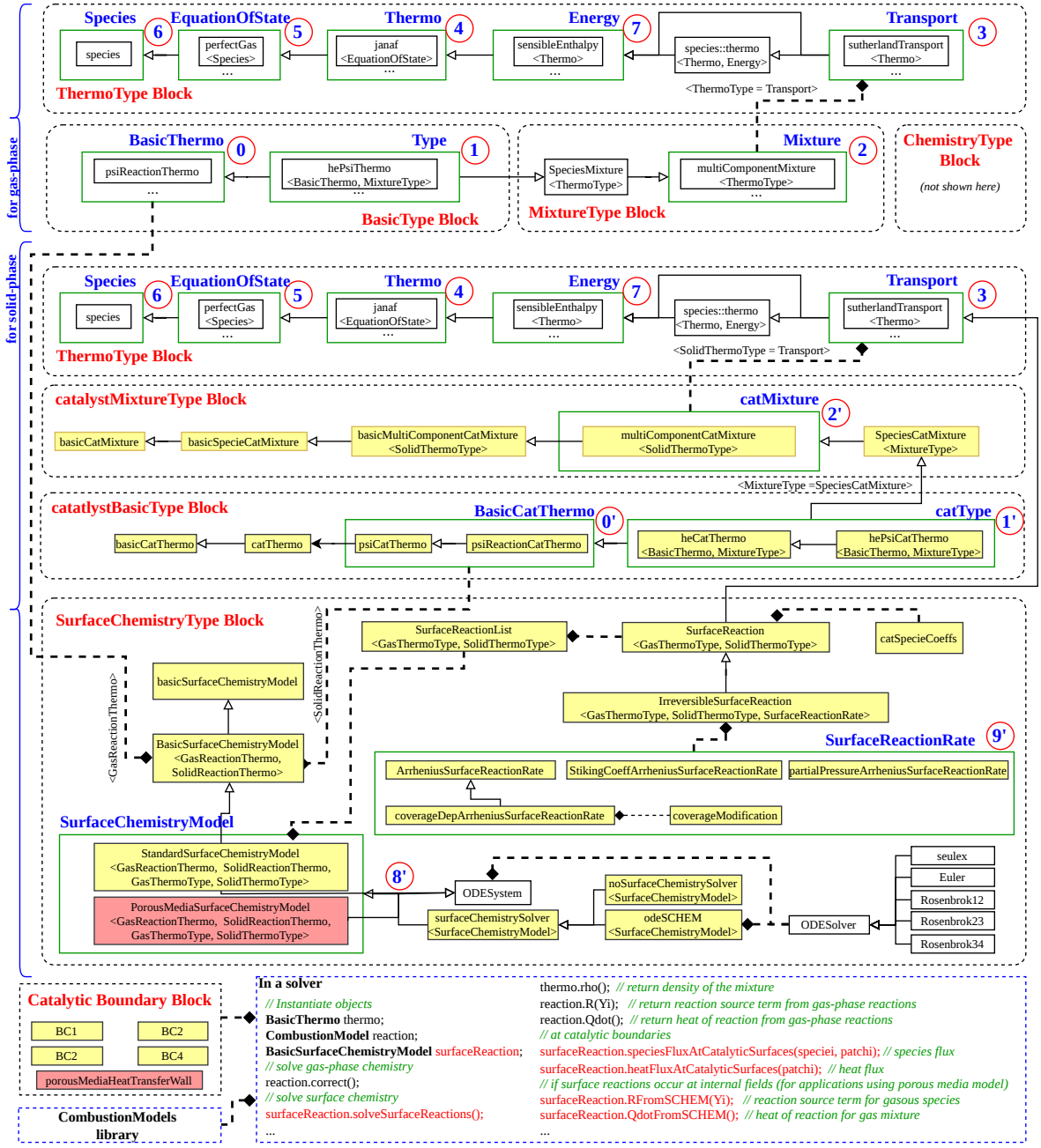


Figure 1: Class diagram of the extended thermophysical library. Yellow boxes denote classes inherited from `surfaceChemistryFoam`, whereas red boxes denote classes introduced in the present work for porous-media modeling. Boxes with circled numbers indicate representative classes selected through the run-time selection mechanism. Solid and dashed arrows indicate inheritance and usage relationships, respectively.

- *porousMediaHeatTransferWall* provides the wall boundary condition for porous-media heat-transfer simulations. It evaluates heat exchange between the reactor wall and packed bed using the heat-transfer coefficient and effective thermal conductivity of the porous medium.

1.2. Solvers and preprocessing utility

Two dedicated OpenFOAM solvers, `porousRhoPBRFoam` and `porousPBRFoam`, were developed for porous-media simulations of packed-bed reactors (PBRs).

The `porousRhoPBRFoam` solver is designed for non-reacting flow and heat-transfer simulations under the constant-density assumption. It solves the macroscopic mass, momentum, energy, and species conservation equations without chemical reactions. It is primarily intended for evaluating pressure drop, heat transfer, and species transport in non-reacting PBRs.

The `porousPBRFoam` solver is designed for reacting-flow simulations with variable gas-mixture properties. It solves the macroscopic mass, momentum, energy, and species conservation equations coupled with detailed heterogeneous surface chemistry. It is primarily intended for evaluating the coupled effects of flow, heat transfer, species transport, and catalytic reactions in PBRs.

For surface-chemistry preprocessing, the framework uses the `surfChemkinToFoam` utility provided by the `surfaceChemistryFoam` library. This utility extends the standard OpenFOAM `chemkinToFoam` utility to convert heterogeneous surface-chemistry input files from CHEMKIN format into OpenFOAM format.

2. Running simulations

2.1. Case-file description for non-reacting flow simulation

The first tutorial, a wall-heated packed-bed case, demonstrates non-reacting flow and heat transfer simulation in a packed-bed reactor (PBR) using the constant-density solver `porousRhoPBRFoam`. The case is available in `tutorials/01_WallHeated_HeatTransfer` within the `porousPBRFoam-8` source repository [2]. It introduces the basic porous-media

modeling (PMM) setup, including the specification of packed-bed properties, porous-media closure models, gas-species properties, and the wall heat-transfer boundary condition for the temperature field.



Figure 2: Case file structure for the non-reacting porous media heat-transfer simulation.

As shown in Fig. 2, the case follows the standard OpenFOAM directory structure, with the addition of the `porousMediaProperties` dictionary, which specifies the packed-bed properties and porous-media closure models.

2.1.1. Settings at time directory

For the non-reacting heat-transfer case, the fields in the initial-time directory are configured in the same manner as those in a standard `reactingFoam` case, except for the wall boundary condition for the temperature field, `T`. This boundary condition accounts for heat transfer between the reactor wall and the packed bed using the effective thermal conductivity of the porous medium.

Setting for the temperature field. For the temperature field `T`, standard boundary conditions such as `fixedValue` and `zeroGradient` can be used in the same manner as in `reactingFoam`.

When a wall-to-bed heat-transfer closure model is used, the reactor-wall boundary condition should be specified as `porousMediaWallHeatTransfer`, as shown below.

```
/*--------------------------*- C++ -*----------------------------*\
  =========                 |
  \\      /  F ield         | OpenFOAM: The Open Source CFD Toolbox
   \\    /   O peration     | Website:  https://openfoam.org
    \\  /    A nd           | Version:  8
     \\/     M anipulation  |
\*---------------------------------------------------------------*/
FoamFile
{
    version     2.0;
    format      ascii;
    class       volScalarField;
    location    "0";
    object      T;
}
// * * * * * * * * * * * * * * * * * * * * * * * * * * * * * * * * * * * * //

dimensions      [0 0 0 1 0 0 0];

internalField   uniform 293;

boundaryField
{
    inlet
    {
        type            fixedValue;
        value           uniform 293;
    }
    wall
    {
        type            porousMediaWallHeatTransfer;
        mode            coefficient;
        kappaMethod     fluidThermo;
        h               uniform 100;
        Ta              constant 373;
        value           $internalField;
    }
    ...
}
// ************************************************************************* //
```

2.1.2. Settings at constant directory

Setting for porousMediaProperties file. The `constant/porousMediaProperties` dictionary specifies solid properties, bed geometry, and closure models for momentum resistance ($\mathbf{S}_m$), thermal conductivity (k_{eff}), species diffusivity ($D_{i,\text{eff}}$), and wall-to-bed heat transfer (h_w), along with model-specific parameters. An example for the non-reacting heat-transfer case is shown below.

```
/*--------------------------*- C++ -*----------------------------*\
```

```
  =========                 |
  \\      /  F ield         | OpenFOAM: The Open Source CFD Toolbox
   \\    /   O peration     | Website:  https://openfoam.org
    \\  /    A nd           | Version:  8
     \\/     M anipulation  |
\*---------------------------------------------------------------------------*/
FoamFile
{
    version     2.0;
    format      ascii;
    class       dictionary;
    location    "constant";
    object      porousMediaProperties;
}
// * * * * * * * * * * * * * * * * * * * * * * * * * * * * * * * * * * * * * //
// Solid Thermophysical properties
SolidThermalConductivity                        1.175;  // [W/m-K]
SolidDensity                                    2719;   // [kg/m3]
SolidHeatCapacity                               871;    // [J/kg-K]

// Porous Media geometric properties
Porosity                                        0.4165; // [-]
ParticleDiameter                                5e-3;   // [m]

// Porous Media closure model
pressureDropClosureModel                        None;
ThermalConductivityClosureModel                 Const;
DiffusivityClosureModel                         BruggemanWilhelm;
WallHeatTransferClosureModel                    Const;

EffectiveThermalConductivity                    0.7;
heatTransferCoefficient                         100;

// ************************************************************************* //
```

The principal entries in the `porousMediaProperties` dictionary, including the solid-phase thermophysical properties and packed-bed geometrical and flow parameters, are described below.

- `SolidThermalConductivity` (k_s) specifies the thermal conductivity of the solid particle.
- `SolidDensity` (ρ_s) specifies the density of the solid particle.
- `SolidHeatCapacity` ($c_{p,s}$) specifies the specific heat capacity of the solid particle.
- `Porosity` (ε) specifies the porosity (void fraction) of the packed bed.
- `ParticleDiameter` (d_p) specifies the diameter of catalytic particle.

The keywords used to select the porous-media closure models are described below.

- `pressureDropClosureModel` selects the momentum-resistance model. Required keywords are:
  - *None*: no momentum resistance.
  - *Darcy*: `DarcyCoeff`.
  - *Forchheimer*: `DarcyCoeff`, `ForchCoeff`.
  - *Ergun*: `Porosity` (ε), `ParticleDiameter` (d_p).
- `DiffusivityClosureModel` supports only the *BruggemanWilhelm* model for evaluating the effective species diffusivity ($D_{i,\text{eff}}$). The required keywords are `ParticleDiameter` and `Porosity`.
- `ThermalConductivityClosureModel` selects the model for evaluating k_{eff}. Available models and their required keywords are:
  - *Const*: `EffectiveThermalConductivity` (k_{eff}).
  - *ConstAnisotropic*: `EffectiveRadialThermalConductivity` (k_r), `EffectiveAxialThermalConductivity` (k_z), `AxialDirection` ($\mathbf{e}_z$).
  - *Standard*: `SolidThermalConductivity` (k_s).
  - *Bunnell*: `SuperficialVelocity` (U_s).
  - *Yagi*: k_s, U_s, `YagiAlphaBeta` ($\alpha\beta$), `YagiPhi` (ϕ).
  - *Wakao*: k_s, U_s, `StagnantThermalConductivityClosureModel` (*Kunii*, *Krupiczka*, *ZS*); *Kunii*: `KuniiPhi` (ϕ), `KuniiGamma` (γ).
- `WallHeatTransferClosureModel` selects the model for h_w. All models require `SolidThermalConductivity` (k_s).
  - *Const*: `heatTransferCoefficient` (h_w).
  - *Yagi*, *Li*, *Bey*: `SuperficialVelocity` (U_s), `ParticleDiameter` (d_p).

  Bey additionally requires `ReactorDiameter` (d_t).

Setting for thermophysicalProperties file. The `thermophysicalProperties` dictionary defines the gas-phase thermophysical models. For `porousRhoPBRFoam`, only the constant-density model combination listed in Table 1 is supported; model names must match those listed exactly. Constant species properties are specified separately in `thermo.STM`.

Table 1: Gas-phase thermophysical-model combination supported by `porousRhoPBRFoam`.

Option	type	mixture	transport	thermo	energy	equationOfState	specie
a	heRhoThermo	multi[2]	const[3]	hConst[4]	senE[5]	rhoConst	specie

where the abbreviations are defined as follows:

multi[2]: `multiComponentMixture`; const[3]: `const` transport;

hConst[4]: constant-C_p thermodynamics; senE[5]: `sensibleEnthalpy`.

Setting the thermo.STM file. The `thermo.STM` file lists the gas species and their constant properties: molecular weight, density, specific heat capacity, formation enthalpy, dynamic viscosity, and Prandtl number. Each entry in the `species` list requires a property block, as illustrated below for O_2.

```
/*--------------------------------*- C++
    -*----------------------------------*
  =========                 |
  \\      /  F ield         | OpenFOAM: The Open Source CFD Toolbox
   \\    /   O peration     | Website:  https://openfoam.org
    \\  /    A nd           | Version:  8
     \\/     M anipulation  |
\*---------------------------------------------------------------------------
    */

FoamFile
{
    version     2.0;
    format      ascii;
    class       dictionary;
    location    "constant";
    object      thermo.STM;
}

species
(
    O2
    H2O
    CH4
    CO2
    N2
);

O2
```

```
{
    specie
    {
        molWeight       31.9988;
    }

    equationOfState
    {
        rho             0.946;
    }

    thermodynamics
    {
        Cp              1008;
        Hf              0;
    }

    transport
    {
        mu              2.174e-05;
        Pr              0.71;
    }
}
```

2.1.3. Settings at system directory

The files located in the `system` directory contain the numerical and simulation-control settings required for running the case. In general, their configuration is identical to that used in `rhoPimpleFoam`, and no special modifications are typically required for catalytic reacting-flow simulations.

2.1.4. Running the non-reacting flow simulations

Navigate to the non-reacting tutorial directory, generate the mesh, and run the solver:

```
$ cd ~/OpenFOAM/yourDirectory/porousPBRFoam-8
$ cd tutorials/01_WallHeated_HeatTransfer
$ blockMesh
$ porousRhoPBRFoam
```

Alternatively, run `./Allrun` from the same directory.

2.2. Case-file description for reacting-flow simulation

The second tutorial case, isothermal NH_3 decomposition, demonstrates reacting flow in a packed-bed reactor using `porousPBRFoam`. It is available in `tutorials/02_Isothermal_`

`NH3Decomposition` in the source repository [2] and corresponds to the isothermal validation case presented in the manuscript.

This case uses `porousPBRFoam` to model variable gas-mixture properties, species transport, and heterogeneous surface chemistry. As shown in Fig. 3, it extends the standard `reactingFoam` case structure with additional fields in the initial-time directory and dictionaries in the `constant` directory for porous-media modeling and surface chemistry.

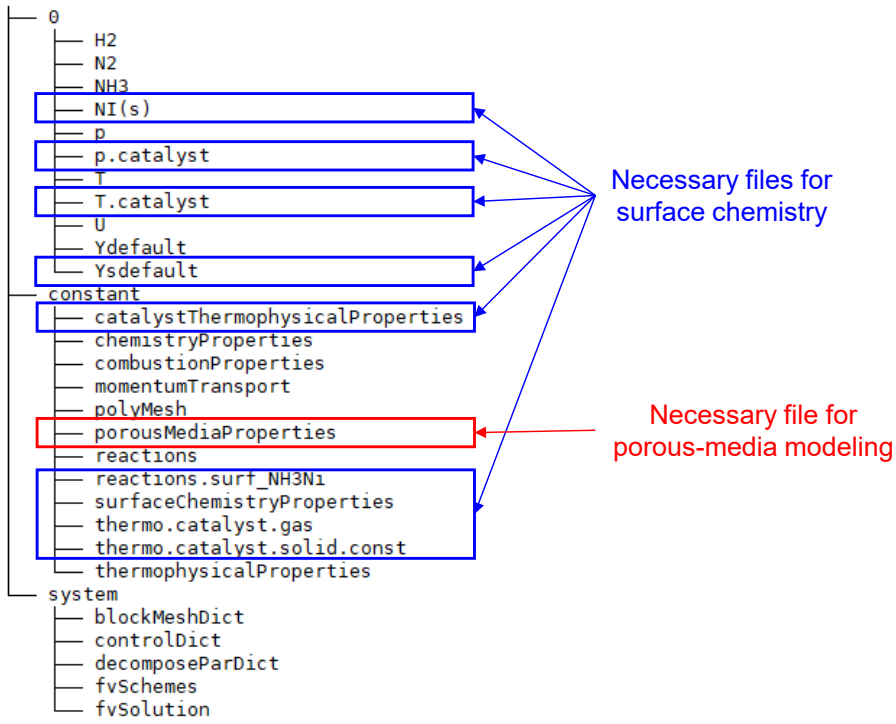


Figure 3: Case-file structure for a reacting-flow simulation using `porousPBRFoam` with heterogeneous surface chemistry. Additional files required for porous-media modeling and heterogeneous surface chemistry are highlighted separately.

In the initial-time directory (e.g., `0`), gas-phase fields are configured as in a standard `reactingFoam` case. Additional files are:

- `p.catalyst`: catalyst pressure.
- `T.catalyst`: catalyst temperature.
- `Ysdefault`: default initial site fractions.

- Individual site-species files (e.g., `NI(s)` and `Ru(s)`): initial fractions of the corresponding surface species.

The required site species depend on the reaction mechanism; intermediate site fractions are initialized to zero unless otherwise specified.

The additional files in `constant` are:

- `porousMediaProperties`: packed-bed properties and closure models.
- `catalystThermophysicalProperties`: catalyst thermophysical models.
- `surfaceChemistryProperties`: surface-chemistry settings.
- `reactions.surf`: heterogeneous reaction mechanism.
- `thermo.catalyst.solid` and `thermo.catalyst.gas`: thermodynamic data for surface and gas species.

2.2.1. Settings at time directory

Surface-species site fractions. Initialize the site fractions to sum to unity, typically by setting the vacant-site species (e.g., `NI(s)`) to 1.0 and adsorbed species to zero. Other initial distributions may be used, but setting all fractions to zero may cause initialization to fail. Standard boundary conditions, such as `zeroGradient`, can be applied.

```
/*---------------------------*- C++ -*-----------------------------*\
  =========                 |
  \\      /  F ield         | OpenFOAM: The Open Source CFD Toolbox
   \\    /   O peration     | Website:  https://openfoam.org
    \\  /    A nd           | Version:  8
     \\/     M anipulation  |
\*-------------------------------------------------------------------*/
FoamFile
{
    version     2.0;
    format      ascii;
    class       volScalarField;
    location    "0";
    object      NI(s);
}
// * * * * * * * * * * * * * * * * * * * * * * * * * * * * * * * * * * * //
dimensions      [0 0 0 0 0 0 0];

internalField   uniform 1;
```

```
boundaryField
{
    heatedWall
    {
        type            zeroGradient;
    }
    inlet
    {
        type            zeroGradient;
    }
    ...
}
// ************************************************************************* //
```

Settings for the temperature fields (T and T.catalyst). The gas-phase temperature field `T` is configured as described in Section 2.1.1. The surface-chemistry library also requires `T.catalyst`, which can be created by copying `T` and setting its boundary conditions to `zeroGradient`. Its values are not used to evaluate heterogeneous reaction rates in the PMM formulation.

Settings for the pressure fields (p and p.catalyst). The gas-phase pressure field `p` is configured as in a standard `reactingFoam` case. The surface-chemistry library also requires `p.catalyst`, which can be created by copying `p` and applying appropriate boundary conditions, such as `zeroGradient`. Its values are not used to evaluate heterogeneous reaction rates in the PMM formulation.

2.2.2. Settings at constant directory

Setting for porousMediaProperties file. Packed-bed properties and closure models are configured as in the non-reacting tutorial. Two additional keywords control heterogeneous surface chemistry and heat transfer in `porousPBRFoam`:

```
/*---------------------------*- C++ -*-----------------------------*\
  =========                 |
  \\      /  F ield         | OpenFOAM: The Open Source CFD Toolbox
   \\    /   O peration     | Website:  https://openfoam.org
    \\  /    A nd           | Version:  8
     \\/     M anipulation  |
\*-------------------------------------------------------------------*/
FoamFile
{
    version     2.0;
    format      ascii;
    class       dictionary;
```

```
    location    "constant";
    object      porousMediaProperties;
}
// * * * * * * * * * * * * * * * * * * * * * * * * * * * * * * * * * * * * //
solverMode                                          PMSCHEM;
isothermal                                          on;

// Solid Thermophysical properties
SolidThermalConductivity                            30;      // [W/m-K]
SolidDensity                                        3540;    // [kg/m3]
SolidHeatCapacity                                   880;     // [J/kg-K]

// Porous Media geometric properties
Porosity                                            0.38;     // [-]
ParticleDiameter                                    2E-5;    // [m]

// Porous Media closure model
pressureDropClosureModel                            Forchheimer;
DiffusivityClosureModel                             BruggemanWilhelm;

// Pressure-drop coefficients
DarcyCoeff                                          1.0508E+07;
ForchCoeff                                          1.9773E+03;

// ************************************************************************* //
```

- `solverMode`: `PM` disables surface chemistry; `PMSCHEM` enables it.
- `isothermal`: `on` maintains the prescribed temperature without solving the energy equation; `off` (default) solves it.

Setting for thermophysicalProperties file. The `thermophysicalProperties` dictionary defines the gas-phase thermophysical models, following the standard `reactingFoam` format. An example is shown in Fig. 4(b), and supported model combinations are listed in Table 2.

a) *catalystThermophysicalProperties* file (for surface-phase)

```
solidThermoType    keyword    Selected model
{
  1' solidType          hePsiCatThermo;
  2' solidMixture       multiComponentCatMixture;
  3' solidTransport     const;
  4' solidThermo        hConst;
  5' solidEnergy        sensibleEnthalpy;
  6' solidEquationOfState rhoConst;
  7' solidSpecie        specie;
}

inertSpecie AR;
#include "thermo.catalyst.solid"
```

b) *thermophysicalProperties* file (for gas-phase)

```
thermoType    keyword    Selected model
{
  1 type          hePsiThermo;
  2 mixture       multiComponentMixture;
  3 transport     sutherland;
  4 thermo        janaf;
  5 energy        sensibleEnthalpy;
  6 equationOfState perfectGas;
  7 specie        specie;
}

inertSpecie AR;
#include "thermo.catalyst.gas"
```

Figure 4: Example settings of the `catalystThermophysicalProperties` file (a) and the `thermophysicalProperties` file (b) for simulations using the `porousPBRFoam` solver.

Table 2: An available combination of gas-phase thermophysical models supported by `porousPBRFoam`

Option	type	mixture	transport	thermo	energy	equationOfState	specie
a	hePsiThermo	multi[a2]	STM[a3]	janaf	senE/senI[5]	perfectGas	specie
b	hePsiThermo	DTMmulti[b2]	DTM[b3]	janaf	senE/senI[5]	perfectGas	specie
c	hePsiThermo	FTMmulti[c2]	FTM[c3]	janaf	senE/senI[5]	perfectGas	specie
d	hePsiThermo	PPmulti[d2]	ppFTM[d3]	janaf	senE/senI[5]	perfectGas	specie

where all abbreviations are:

senE/senI[5]:`sensibleEnthalpy`/`sensibleInternalEnergy`;

multi[a2]: `multiComponentMixture`; STM[a3]: `sutherland`;

DTMmulti[b2]: `DTMMultiComponentMixture`; DTM[b3]: `DTMTransport`;

FTMmulti[c2]: `FTMMultiComponentMixture`; FTM[c3]: `FTMTransport`;

PPmulti[d2]: `DTMMultiComponentMixture`; ppFTM[d3]: `preProcessingFTMTransport`;

Combination (a) uses standard transport with unity Lewis and Schmidt numbers. Combination (b) uses the detailed transport model (DTM) and requires `usingDetailedTransportModel` to be `true` (default: `false`). Combination (c) uses the fitting transport model (FTM) of Nguyen et al. [3], which reduces the cost of detailed transport. Combination (d) generates FTM input data during preprocessing.

Instructions for DTM, the Soret effect, FTM, and coTHERM are provided in `DTLReactingFoam_userGuide_OF8_updated.pdf` under `surfaceChemistryFoam-8/documentations` [4].

Setting for catalystThermophysicalProperties file. The The `catalystThermophysicalProperties` dictionary defines the thermophysical models for the catalyst phase and surface species. Model names must match the supported configuration shown in Fig. 4(a).

Setting for chemistryProperties file. The `chemistryProperties` dictionary is configured identically to that used in a standard `reactingFoam` case.

Setting for surfaceChemistryProperties file. The `surfaceChemistryProperties` dictionary defines the surface-chemistry model and integration settings. Reaction rates in the porous region are converted from surface-area-based rates to volumetric source terms using the specified surface-to-volume ratio.

```
/*--------------------------*- C++ -*-----------------------------*\
  =========                 |
  \\      /  F ield         | OpenFOAM: The Open Source CFD Toolbox
   \\    /   O peration     | Website:  https://openfoam.org
    \\  /    A nd           | Version:  8
     \\/     M anipulation  |
\*-------------------------------------------------------------------*/
FoamFile
{
    version     2.0;
    format      ascii;
    class       dictionary;
    location    "constant";
    object      surfaceChemistryProperties;
}
// * * * * * * * * * * * * * * * * * * * * * * * * * * * * * * * * * * * * //

// Select the porous-media surface-chemistry treatment
onlySurfaceCatalyst false; // default: false

// Compatibility entries; leave empty for the present PMM setup
catalystRegions
{
    catalyticSurfaces   ( );
    catalyticZones      ( );
}

// Catalytic surface area per unit reactor volume [1/m]
SurfaceToVolumeRatio            2.8E5;

// Surface-chemistry method and ODE solver
surfaceChemistryType
{
    method          porousMediaSCHEM;   // porous-media model
    solver          odeSCHEM;
}

chemistry           on;

initialChemicalTimeStep 1e-07;

odeSCHEMCoeffs
{
    solver          seulex;
    absTol          1e-8;
    relTol          0.01;
}

#include "reactions.surf"
// ************************************************************************ //
```

The main entries in `surfaceChemistryProperties` are:

- `onlySurfaceCatalyst`: set to `false` for PMM.
- `catalystRegions`: retained for library compatibility; leave `catalyticSurfaces` and

`catalyticZones` empty.

- `SurfaceToVolumeRatio`: catalytic interfacial area per unit porous-region volume, used to convert surface species-production and heat-release rates into volumetric source terms.
- `surfaceChemistryType`: set `method` to *porousMediaSCHEM* and `solver` to *odeSCHEM*, the only supported solver.
- `odeSCHEMCoeffs`: ODE integrator and tolerances; the example uses `seulex` with absolute and relative tolerances of 10^{-8} and 10^{-2}.
- `chemistry`: enables or disables surface chemistry.
- `initialChemicalTimeStep`: initial time step for chemistry ODE integration.

The surface reaction mechanism is included using `#include"reactions.surf"`. The `surfChemkinToFoam` utility can be used to prepare the required OpenFOAM input files. For example, the Appari NH_3 decomposition tutorial uses:

```
surfChemkinToFoam
    surf.inp
    surfTherm.dat
    surfaceThermoDict
    reactions.surf
    thermo.catalyst.solid.const
```

Detailed preparation instructions are provided in `prepareReactionsSurfFile.pdf` under `surfaceChemistryFoam-8/documentations` [4].

2.2.3. Settings at system directory

The files located in the `system` directory contain the numerical and simulation-control settings required for running the case. In general, their configuration is identical to that used in `reactingFoam`, and no special modifications are typically required for catalytic reacting-flow simulations.

2.2.4. Running the reacting flow simulations

Navigate to the reacting-flow tutorial directory, generate the mesh, and run the solver:

```
$ cd ~/OpenFOAM/yourDirectory/porousPBRFoam-8
$ cd tutorials/02_Isothermal_NH3Decomposition
$ blockMesh
$ porousPBRFoam
```

Alternatively, run `./Allrun` from the same directory. These tutorials provide a starting point for configuring and running other PBR simulations.

References

[1] D. N. Nguyen, J. H. Lee, H. W. Seo, H. J. Ahn, C. S. Yoo, surfaceChemistryFoam: An OpenFOAM-based library for detailed surface chemistry in reacting flow simulations, Comput. Phys. Commun. 327 (2026) 110290.

[2] `https://github.com/danhnam11/porousPBRFoam-8`.

[3] D. N. Nguyen, J. H. Lee, C. S. Yoo, DTLreactingFoam: An efficient CFD tool for laminar reacting flow simulations using detailed chemistry and transport with time-correlated thermophysical properties, Comput. Phys. Commun. 322 (2026) 110052.

[4] `https://github.com/danhnam11/surfaceChemistryFoam-8`.